Chapter 4 — Linear Integrated Circuits

Syllabus (Annexure-VII C, Unit 4): Introduction to Operational Amplifier · Linear applications · Non-linear applications · Voltage regulators · Timers · Phase-Locked Loop

📊 Weightage & Trend Box

Item	Detail
Direct weight (22-04-2025 CBT-2, Shift-2)	**2 / 50 technical Q ≈ 4%** (scales to ~4/100 in full technical section)
Q1 observed	**IC 741 internal stages** — "Which is the *second* (intermediate) stage of IC 741?" → **Intermediate / gain stage** (also described as level-shift + gain stage in some texts)
Q2 observed	**Op-amp integrator** — output is **proportional to the integral of the input** ($V_o = -(1/RC)\int V_{in}\,dt$)
Hidden weight	**PLL** questions appeared under **Communication Engineering** (FM demodulation, AM detection by PLL). So PLL preparation here pays off in the 16%-weight Communication unit too.
Trend	Questions are **conceptual/one-step numerical**: op-amp golden rules, standard gain formulas, 555 timing formulas, 78xx/LM317 identification, PLL block roles, capture vs lock range. Nothing beyond standard formulas has been asked.
Strategy	Memorise the **formula sheet** at the end cold; be able to *draw mentally* the 741 stage order, 555 internal block, and PLL block diagram.

4.1 Integrated Circuit (IC) Basics

An **Integrated Circuit** fabricates an entire circuit (transistors, diodes, resistors, small capacitors) on a single chip of semiconductor (usually silicon).

Classification by fabrication:

Type	Description	Remarks
Monolithic IC	All components fabricated on a *single* silicon chip by diffusion/ion implantation	Cheapest in mass production; IC 741, 555, 7400 are monolithic
Hybrid IC	Separate chips + thin/thick-film passive components interconnected on one insulating substrate	Used for high power / high precision
Thin-film / Thick-film	Passive components deposited as films on glass/ceramic; active devices added separately	Film thickness: thin < 1 µm; thick > 1 µm (screen-printed)

Classification by signal: Linear (analog) ICs — output varies continuously with input (op-amps, regulators, timers in linear mode); **Digital** ICs — two-state output (gates, flip-flops).

Classification by integration level: SSI (< 12 gates), MSI (12–99), LSI (100–9999), VLSI (10^4–10^5), ULSI (> 10^5 gates/components).

Advantages of ICs: small size & weight, low cost in volume, high reliability (fewer solder joints), low power, matched components (excellent for differential pairs), improved speed (short interconnects).

Limitations: inductors and large capacitors cannot be fabricated, limited power dissipation, low voltage ratings, cannot be repaired (replace only).

4.1.1 IC 741 — the classic op-amp

- **Monolithic**, general-purpose, internally frequency-compensated (30 pF on-chip capacitor → no external compensation needed), **8-pin DIP**.
- Supply: typically **±15 V** (range ±5 V to ±18 V).

IC 741 pin diagram (8-pin DIP, top view):

Pin	Function	Pin	Function
1	Offset null	8	NC (no connection)
2	Inverting input (−)	7	+V_{CC}
3	Non-inverting input (+)	6	Output
4	−V_{EE}	5	Offset null

Internal block diagram — four stages in order (★ asked in 2025):

```
Input → [1. Input differential amplifier stage]
      → [2. Intermediate / gain stage]          ← ★ SECOND stage (2025 PYQ)
      → [3. Level-shifting (level translator) stage]
      → [4. Output (push-pull class-AB) stage] → Output
```

Stage	Circuit used	Job
1. Input stage	**Dual-input, balanced-output differential amplifier**	High input resistance, high CMRR, most of the voltage gain begins here
2. Intermediate (gain) stage	**Dual-input, unbalanced (single-ended) output differential amplifier**	Provides the bulk of additional voltage gain
3. Level shifter	**Emitter follower with level translator** (V_{BE}-shift network / current source)	Shifts the DC level down so output can swing symmetrically about 0 V (direct coupling accumulates DC)
4. Output stage	**Complementary-symmetry push-pull (class AB)**	Low output resistance, high current drive, large voltage swing

> **Exam nuance:** Some books club stages 2 and 3 and call the second stage the "**level-shift and gain stage**." The 2025 paper's accepted answer for "second/intermediate stage of 741" was the **intermediate gain stage** — remember both names.

Typical 741 data: open-loop gain A ≈ 2×10^5 (≈ 106 dB), input resistance ≈ 2 MΩ, output resistance ≈ 75 Ω, CMRR ≈ 90 dB, slew rate ≈ **0.5 V/µs**, unity-gain bandwidth ≈ 1 MHz, input offset voltage ≈ 2 mV, input bias current ≈ 80 nA, input offset current ≈ 20 nA.

4.2 Op-Amp Fundamentals

An **operational amplifier** is a very-high-gain, direct-coupled differential amplifier. Name comes from its original use for mathematical *operations* (add, subtract, integrate, differentiate) in analog computers.

Symbol: triangle with two inputs — inverting (−) and non-inverting (+) — and one output.

Output (open loop): **$V_0 = A \cdot (V_+ - V_-)$** where A = open-loop differential gain.

4.2.1 Ideal vs practical (741) characteristics ★

Parameter	Ideal op-amp	Practical (IC 741)
Open-loop voltage gain A	∞	≈ 2×10^5
Input resistance R_{in}	∞	≈ 2 MΩ
Output resistance R_{out}	0	≈ 75 Ω
Bandwidth	∞	≈ 1 MHz (unity-gain)
CMRR	∞	≈ 90 dB
Slew rate	∞	≈ 0.5 V/µs
Input offset voltage/current	0	≈ 2 mV / 20 nA
Output for $V_+ = V_-$ (common-mode gain)	0	small but non-zero
Drift with temperature	0	present

Key definitions:

- **CMRR** = |A_d/A_{cm}|; in dB: CMRR = 20 log(A_d/A_{cm}). Higher CMRR ⇒ better rejection of common-mode noise (hum picked up equally on both wires).
- **Slew rate (SR)** = maximum rate of change of output voltage = $(dV_0/dt)_{max}$, in V/µs. For an undistorted sine output of peak V_m at frequency f: required SR ≥ 2πf·V_m. Max full-power frequency **$f_{max} = SR/(2\pi V_m)$**.
- **Gain–bandwidth product (GBW):** for a single-pole op-amp, $A_{CL} \times f_{CL}$ = constant = unity-gain bandwidth (1 MHz for 741). Closed-loop bandwidth = GBW / closed-loop gain.
- **Input offset voltage V_{io}:** DC voltage that must be applied between inputs to force $V_0 = 0$. Nulled in 741 using a 10 kΩ pot between pins 1 and 5 (wiper to −V_{EE}).
- **PSRR:** change in input offset voltage per unit change in supply voltage (µV/V).

4.2.2 Golden rules, virtual short, virtual ground ★

With **negative feedback** and very large A:

1. **Rule 1 (virtual short):** $V_+ \approx V_-$ — the two input terminals sit at (virtually) the same potential, because V_0 is finite and $V_0/A \to 0$.
2. **Rule 2:** No current enters either input terminal ($R_{in} \to \infty$).

Virtual ground: In the *inverting amplifier* the (+) input is grounded, so by Rule 1 the (−) input is also at ≈ 0 V *without being physically connected to ground* — it is a **virtual ground**: zero voltage, but it can sink/source signal current through the feedback path.

> Virtual short exists **only with negative feedback and the op-amp in its linear region**. In comparators/Schmitt triggers (no negative feedback), there is **no** virtual short.

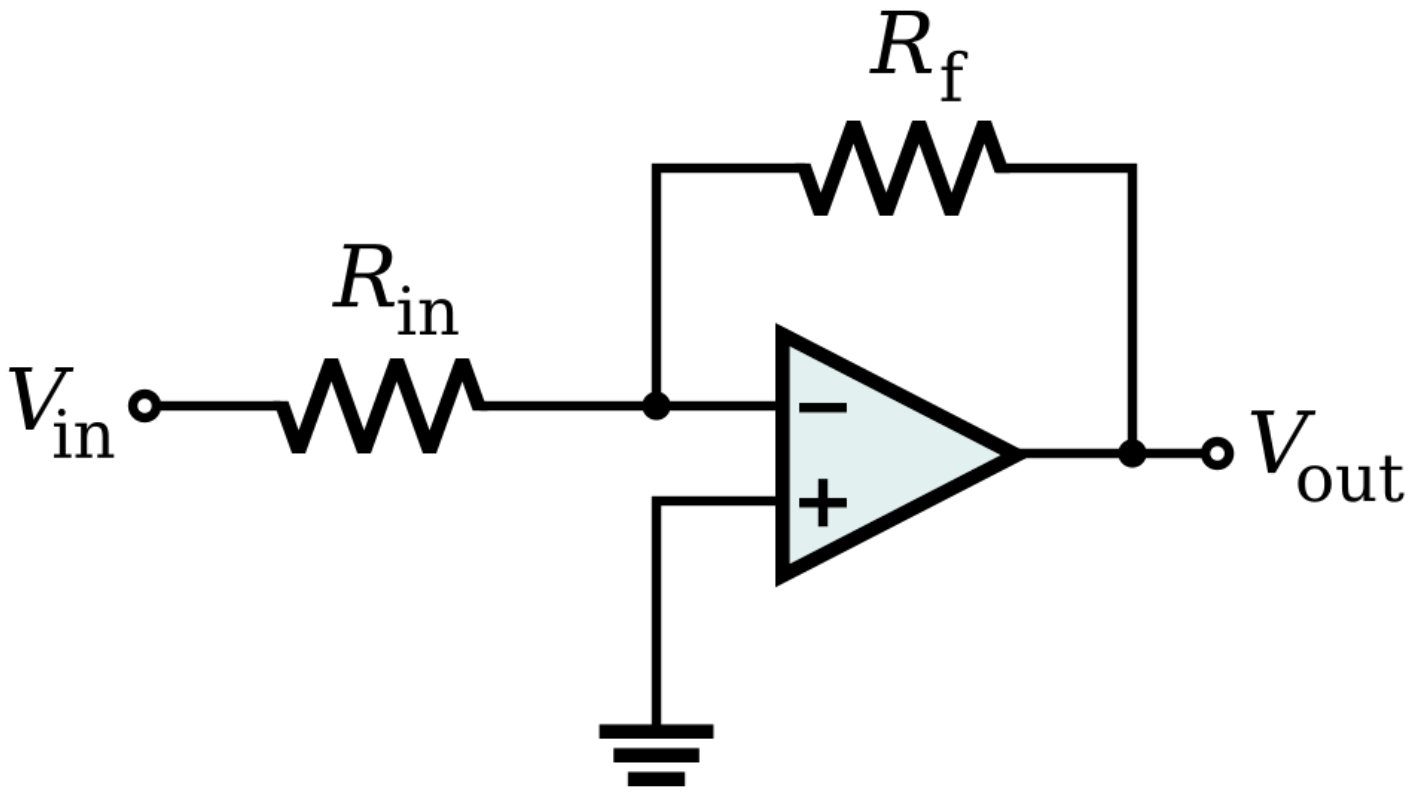
4.2.3 Open-loop configurations

No feedback ⇒ gain ≈ 2×10^5 ⇒ even a few tens of µV drives the output to saturation (±V_{sat} ≈ ±13 V with ±15 V supplies). Three open-loop modes: differential amplifier, inverting amplifier, non-inverting amplifier — all impractical as linear amplifiers (output slams to ±V_{sat}); the useful open-loop application is the **comparator** (§4.4.1).

4.3 Linear Applications of Op-Amps

"Linear" = output is a linear (proportional / weighted-sum / calculus) function of the input; negative feedback keeps the op-amp in its active region.

4.3.1 Inverting amplifier ★



Inverting amplifier (gain −Rf/Rin).

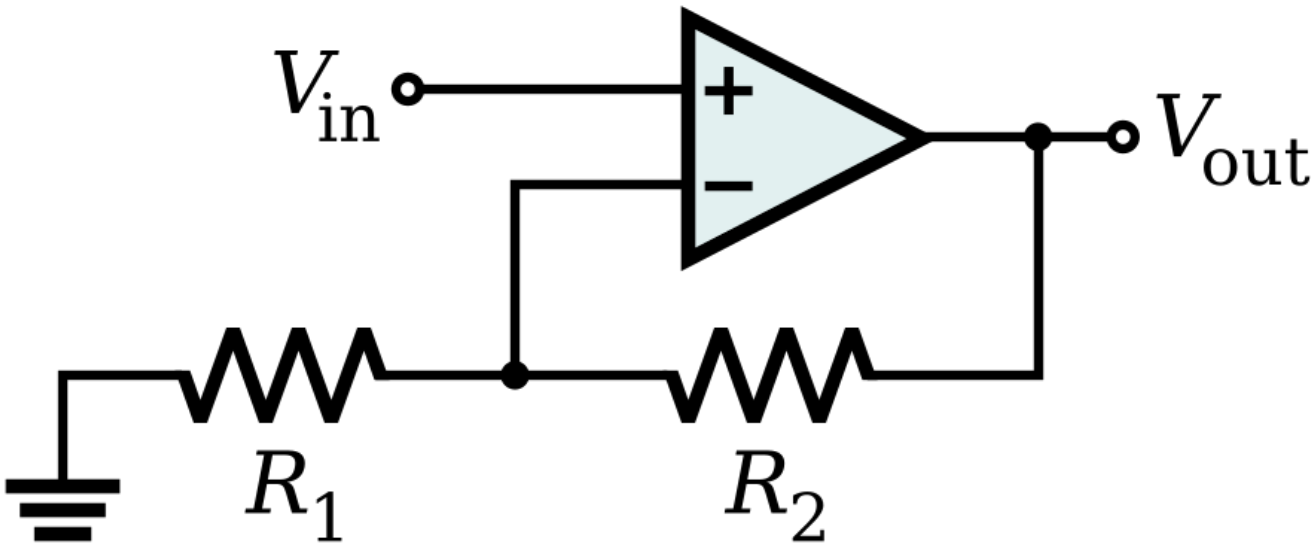
Input via R_{in} (often written R_1) to the (−) terminal; feedback R_f from output to (−); (+) grounded.

Derivation: (−) node is virtual ground ⇒ current through R_1: I = V_{in}/R_1. No current enters op-amp ⇒ same I flows through R_f: 0 − V_0 = I·R_f.

$A_v = V_0/V_{in} = -R_f/R_1$ (180° phase inversion)

Input impedance = R_1 (moderate); output impedance ≈ 0.

4.3.2 Non-inverting amplifier ★



Non-inverting amplifier (gain 1+Rf/Rin).

Signal to (+); feedback divider R_f, R_1 from output to (−) to ground.

Virtual short ⇒ V_- = V_{in}; divider: $V_- = V_0 \cdot R_1/(R_1+R_f)$.

$A_v = 1 + R_f/R_1$ (no phase inversion; gain always ≥ 1)

Very high input impedance (signal sees op-amp's (+) input directly).

4.3.3 Voltage follower (unity-gain buffer)

Non-inverting with $R_f = 0$ (direct wire), $R_1 = \infty$ ⇒ **$A_v = +1$**. Extremely high Z_{in}, near-zero Z_{out} → used for **impedance matching / buffering** (isolating a high-impedance source from a low-impedance load). Best bandwidth of all configurations (BW = full GBW).

4.3.4 Summing amplifier (adder)

Inverting summer: inputs V_1, V_2, V_3 through R_1, R_2, R_3 into the virtual-ground node, feedback R_f:

$V_0 = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$

If $R_1 = R_2 = R_3 = R_f$: $V_0 = -(V_1+V_2+V_3)$. If $R_f = R/3$ with equal Rs: **averager** $V_0 = -(V_1+V_2+V_3)/3$. Weighted summing is the basis of a simple **DAC** (binary-weighted resistor DAC).

Non-inverting summer (two equal inputs): $V_+ = (V_1+V_2)/2$; with gain $(1+R_f/R_1) = 2$ ⇒ $V_0 = V_1+V_2$.

4.3.5 Difference (subtractor) amplifier

Four resistors: V_1 through R_1 to (−) with feedback R_2; V_2 through R_1 to (+) with R_2 to ground (matched pairs).

$V_0 = (R_2/R_1)(V_2 - V_1)$

Rejects common-mode signals *if resistors match perfectly*; mismatch degrades CMRR — hence the instrumentation amplifier.

4.3.6 Instrumentation amplifier

Three op-amps: two non-inverting input buffers sharing gain-set resistor R_G, followed by a difference amplifier.

$V_0 = (1 + 2R/R_G) \cdot (R_3/R_2) \cdot (V_2 - V_1)$ (commonly with $R_3 = R_2$: $V_0 = (1 + 2R/R_G)(V_2 - V_1)$)

Properties: **very high input impedance on both inputs, very high CMRR, gain set by a single resistor R_G**. Used for transducer/bridge signals (strain gauge, thermocouple, biomedical ECG). IC example: AD620.

4.3.7 Integrator ★ (2025 PYQ)

Inverting configuration with feedback **capacitor C** replacing R_f; input through R.

Derivation: virtual ground ⇒ $I = V_{in}/R$; this charges C: $I = -C \cdot dV_0/dt$.

$V_0(t) = -(1/RC) \int_0^t V_{in}\, dt + V_0(0)$

> ★ **Output is proportional to the (negative) integral of the input** — exact 2025 statement.

Input–output waveform pairs (memorise):

Input	Integrator output
Step (DC level)	Negative-going **ramp**
Square wave	**Triangular wave**
Triangular wave	Parabolic segments
Sine wave	**−Cosine** (i.e. sine shifted; gain falls with f)

As a filter, the ideal integrator gain = 1/(ωRC): gain **∞ at DC** (any input offset integrates to saturation — the practical problem) and rolls off at **−20 dB/decade**; phase = +90° (output lags input appropriately as cos → −sin).

Practical (lossy) integrator: a resistor R_f (typically ≈ 10R) is placed **in parallel with C** to give finite DC gain ($-R_f/R$) and prevent saturation. It integrates properly only for **$f \gg f_a = 1/(2\pi R_f C)$**; useful range roughly f_a to $f_b = 1/(2\pi RC)$ (f_b = 0-dB frequency of the ideal response).

4.3.8 Differentiator

Swap R and C of the integrator: input through **C**, feedback **R**.

$I = C \cdot dV_{in}/dt$ through C ⇒ **$V_0 = -RC \cdot (dV_{in}/dt)$**

Input	Differentiator output
Triangular wave	**Square wave**
Square wave	Sharp **spikes/impulses** at edges
Sine	−Cosine (gain rises with f)

Gain = ωRC rises at **+20 dB/decade** ⇒ amplifies high-frequency **noise** and can be unstable. **Practical differentiator:** add series resistor R_1 with the input capacitor (and often a small C_f across R) to limit high-frequency gain; differentiates properly for $f \ll 1/(2\pi R_1 C)$.

4.3.9 V-to-I and I-to-V converters

- **Current-to-voltage (transresistance) converter:** input current I_{in} into the virtual-ground (−) node, feedback R_f ⇒ **$V_0 = -I_{in} R_f$**. Used to read photodiode/photocell currents, DAC output currents.

- **Voltage-to-current (transconductance) converter:** load placed in the feedback path (floating load): **I_L = V_{in}/R₁ — independent of load resistance**. Grounded-load version = **Howland current source**. Used to drive LEDs, deflection coils, 4–20 mA loops.

Worked Examples — linear applications

WE-1 (Inverting gain). R_1 = 10 kΩ, R_f = 100 kΩ, V_{in} = 0.5 V sine peak.

A = −R_f/R_1 = −10 ⇒ output = 5 V peak, phase-inverted. Input impedance = 10 kΩ.

WE-2 (Non-inverting). R_1 = 2 kΩ, R_f = 18 kΩ ⇒ A = 1 + 18/2 = **10**. If V_{in} = 200 mV ⇒ V_0 = 2 V, in phase. Closed-loop bandwidth with a 741 (GBW = 1 MHz): BW = 1 MHz/10 = **100 kHz**.

WE-3 (Summer). R_1 = R_2 = 10 kΩ, R_f = 20 kΩ, V_1 = +1 V, V_2 = −0.5 V.

V_0 = −20k(1/10k − 0.5/10k) = −20k × (0.05 mA) = **−1 V**.

WE-4 (Integrator, step input) ★ pattern of 2025 PYQ. R = 100 kΩ, C = 1 µF, V_{in} = +2 V step at t = 0, $V_0(0)$ = 0.

$V_0(t)$ = −(1/RC)∫2 dt = −(2/0.1)·t = **−20t volts** — a ramp falling at 20 V/s. With ±V_{sat} = ±13 V it saturates at t = 13/20 = **0.65 s**.

WE-5 (Integrator, square input). ±5 V square wave, 1 kHz (T = 1 ms), into R = 10 kΩ, C = 0.1 µF (RC = 1 ms).

During each half-cycle (0.5 ms) the output ramps by ΔV = (1/RC)·V·(T/2) = (1/1 ms)(5)(0.5 ms) = 2.5 V ⇒ **triangular wave of 2.5 V peak-to-peak** (centred, inverted slope sense).

WE-6 (Slew-rate limit). A 741 (SR = 0.5 V/µs) must output a 10 V-peak sine. Max frequency:

f_{max} = SR/(2πV_m) = 0.5×10^6 / (2π×10) ≈ **7.96 kHz**. Above this the sine distorts into a triangle.

4.4 Non-Linear Applications of Op-Amps

Op-amp deliberately driven into saturation (no/positive feedback) or with non-linear feedback elements (diodes).

4.4.1 Comparator & zero-crossing detector

Open-loop op-amp: V_0 = +V_{sat} if $V_+ > V_-$, −V_{sat} if $V_+ < V_-$.

- **Zero-crossing detector:** reference = 0 V; a sine input yields a **square wave** output that switches at every zero crossing. (Basis of frequency measurement, sine-to-square conversion.)
- Non-zero reference V_{ref} on one input ⇒ level detector.
- Weakness: **noise near the threshold causes multiple false transitions (chatter)** — solved by the Schmitt trigger.
- Dedicated comparator ICs (LM311, LM339) have faster response and open-collector outputs.

4.4.2 Schmitt trigger (regenerative comparator) ★

Comparator with **positive feedback**: output fed back to (+) input through divider R_1 (to ground) and R_2 (to output); signal applied to (−).

Two trip points (with symmetric saturation ±V_{sat} and reference ground):

UTP = +V_{sat}·R_1/(R_1+R_2) LTP = −V_{sat}·R_1/(R_1+R_2)

Hysteresis width V_H = UTP − LTP = 2V_{sat}·R_1/(R_1+R_2)

- Rising input must exceed UTP to switch output low; falling input must drop below LTP to switch high (inverting Schmitt). The transfer characteristic is a **hysteresis loop**.
- Noise smaller than V_H cannot cause false triggering — the key advantage over a plain comparator.
- Applications: squaring circuit, noise-immune level detection, relaxation oscillators, debouncing.

4.4.3 Precision rectifiers (super-diode)

An ordinary diode cannot rectify signals below its ~0.7 V drop. Placing the diode **inside the op-amp's feedback loop** divides the effective diode drop by open-loop gain (0.7/A ≈ μV) ⇒ rectification of millivolt signals.

- **Precision half-wave:** op-amp + diode in feedback; output follows input for one polarity, 0 for the other.
- **Precision full-wave (absolute-value circuit):** half-wave stage + summing stage ⇒ $V_0 = |V_{in}|$. Used in AC voltmeters/measurement.

4.4.4 Log and antilog amplifiers (brief)

- **Log amp:** feedback element = diode or transistor (BE junction). Since I = I_se^(V/V_T): **$V_0 = -V_T \cdot \ln(V_{in}/(I_s R))$** — output ∝ logarithm of input.
- **Antilog (exponential) amp:** diode/transistor at the *input*, R in feedback ⇒ output ∝ e^(V_{in}/V_T).
- Cascading log → summer → antilog gives **analog multiplication/division**; also used for dB displays and signal compression.

4.4.5 Op-amp clippers and clampers

- **Active positive/negative clipper:** op-amp with diode + reference V_{ref} clips the waveform precisely at V_{ref} (no 0.7 V error, unlike passive diode clippers).
- **Active clamper (DC restorer):** capacitor in series with input, diode in feedback — shifts the whole waveform so its peak sits at V_{ref}; adds a precise DC level to a signal.
- **Peak detector:** precision half-wave rectifier charging a hold capacitor — output holds the input's peak value.

4.4.6 Waveform generators using op-amps

(a) Square-wave generator (op-amp astable / relaxation oscillator): Schmitt trigger + RC timing network from output to (−) input. The capacitor charges toward ±V_{sat} and the Schmitt flips each time it crosses ±βV_{sat} ($\beta = R_1/(R_1+R_2)$).

$T = 2RC\cdot\ln((1+\beta)/(1-\beta))$; if $R_1 = R_2$ (β = 0.5): **$T \approx 2.2\,RC$**, $f \approx 1/(2.2RC)$, 50% duty cycle.

(b) Triangular-wave generator: square-wave source (Schmitt/comparator) → **integrator** → triangular output; triangle fed back to the comparator closes the loop. Output frequency **$f = R_2/(4R_1 \cdot R\,C)$** for the standard two-op-amp circuit (comparator with divider R_1–R_2 + integrator RC); triangle amplitude = V_{sat}·R_1/R_2. Key idea for exams: **square in → integrator → triangle out**.

(c) Wien-bridge oscillator (sine): op-amp non-inverting amplifier + Wien network (series RC in series with parallel RC) as positive feedback.

$f_0 = 1/(2\pi RC)$; the Wien network's feedback fraction at f_0 = **1/3**, so the amplifier must give gain ≥ 3 ⇒ **R_f = 2R₁** ($1 + R_f/R_1 = 3$). Amplitude stabilised by a thermistor/lamp or diode network. Standard audio-frequency sine source; low distortion.

(Phase-shift oscillator with op-amp: $f_0 = 1/(2\pi RC\sqrt{6})$, required gain 29 — covered fully in the Devices chapter's oscillator section.)

4.4.7 Active filters ★

Active filter = RC network + op-amp: no inductors, provides gain, buffering, easy tuning. (Passive R-L-C filters have no gain and suffer loading.)

Filter	1st-order roll-off	2nd-order roll-off	Cut-off / centre formula
Low-pass (LP)	−20 dB/decade (−6 dB/octave)	−40 dB/decade	f_H = 1/(2πRC) (equal R, C: f = 1/(2πRC))
High-pass (HP)	+20 dB/dec below f_L	+40 dB/dec	f_L = 1/(2πRC)
Band-pass (BP)	—	—	f_0 = √(f_Lf_H); BW = f_H − f_L; Q = f_0/BW
Band-reject (BR/notch)	—	—	Twin-T notch: f_N = 1/(2πRC)

- **First-order LP:** RC to (+) input, op-amp with gain A_F = 1+R_f/R_1; response: |A| = A_F/√(1+(f/f_H)^2); at f = f_H gain is down **3 dB** (0.707A_F).
- **Second-order (Sallen–Key):** two RC sections; equal-component: f_H = 1/(2πRC).
- **Butterworth:** *maximally flat* passband; a 2nd-order Butterworth Sallen–Key needs pass-band gain **1.586** (damping ζ = 0.707, Q = 0.707). Chebyshev = steeper roll-off but ripple; Bessel = best (linear) phase.
- **Wide-band BP** = HP cascaded with LP (f_L < f_H). **Notch** = LP + HP summed, or twin-T; classic use: removing 50 Hz hum.
- **All-pass filter:** flat gain, only phase varies with f — used for phase equalisation/delay.

WE-7 (Filter design). First-order Butterworth LP with f_H = 1 kHz, C = 0.1 µF chosen:

R = 1/(2πf_HC) = 1/(2π×10^3×10^{-7}) ≈ **1.59 kΩ**. For pass-band gain 2, choose R_f = R_1 = 10 kΩ.

4.5 Voltage Regulators

A voltage regulator keeps the DC output constant against **line (input) variation, load variation and temperature**.

- **Line regulation** = $\Delta V_0/\Delta V_{in}$ (output change per volt of input change).
- **Load regulation** = (V_{NL} − V_{FL})/V_{FL} × 100%.

4.5.1 Shunt vs series (linear) regulators

	Shunt regulator	Series regulator
Control element	In **parallel** with load (e.g. zener + R)	In **series** with load (pass transistor)
Simplest example	Zener shunt regulator	Emitter-follower with zener reference
Efficiency	Poor (draws current even at no load)	Better; standard for linear ICs
Short-circuit behaviour	Inherently short-circuit tolerant (series R limits)	Needs added current limiting

A full series regulator = **reference (zener) + error amplifier (op-amp) + series pass transistor + sampling divider** — negative feedback holds V_0 = V_{ref}(1 + R_1/R_2).

4.5.2 Three-terminal fixed IC regulators — 78xx / 79xx ★

Series	Polarity	Common members	Pin order (TO-220, front view)
78xx	**Positive**	7805 (+5 V), 7809, 7812, 7815	1 = Input, 2 = Ground, 3 = Output
79xx	**Negative**	7905 (−5 V), 7912, 7915	1 = **Ground**, 2 = Input, 3 = Output (note the different order!)

- "xx" = output voltage. Typical output current up to 1 A (with heat sink); 78Lxx = 100 mA versions.
- **Dropout voltage ≈ 2 V**: input must exceed output by ≥ ~2 V (e.g. 7805 needs V_{in} ≥ ~7 V). LDO (low-dropout) regulators work with ~0.1–0.6 V headroom.
- Built-in **thermal shutdown, current limiting and SOA protection**.
- 0.33 µF input and 0.1 µF output capacitors recommended for stability.
- To make 78xx adjustable/boosted: add a resistor divider or zener in the ground leg — V_0 = V_{xx} + V_{ground-leg}.

4.5.3 IC 723 — general-purpose adjustable regulator

- 14-pin IC containing **temperature-compensated reference (≈ 7.15 V), error amplifier, series pass transistor (150 mA) and current-limit transistor**.
- Output adjustable **2 V – 37 V**; both **low-voltage (V_0 < 7.15 V, via divider on the reference)** and **high-voltage (V_0 > 7.15 V, via divider in feedback)** connections; external pass transistor extends current to amps; supports **foldback current limiting**.

4.5.4 LM317 — adjustable positive regulator ★

Three terminals: **Input, Output, Adjust**. It maintains **1.25 V (V_{ref}) between Output and Adjust**; a constant ≈ 50–100 µA flows from ADJ (usually neglected).

$V_0 = 1.25\,(1 + R_2/R_1)$ (+ I_{ADJ}R_2, usually neglected)

R_1 (Output→Adjust) is typically 240 Ω. Range 1.25–37 V, up to 1.5 A. Negative counterpart: **LM337**.

WE-8 (LM317). R_1 = 240 Ω, R_2 = 2.4 kΩ:

$V_0 = 1.25(1 + 2400/240) = 1.25 \times 11 =$ **13.75 V**.

For V_0 = 5 V with R_1 = 240 Ω: $R_2 = R_1(V_0/1.25 - 1) = 240(4-1) =$ **720 Ω**.

4.5.5 Linear vs switching (SMPS) regulators

Feature	Linear (series)	SMPS
Pass element operation	Active (rheostat) region — continuous	Switch (cut-off/saturation) at 20 kHz–1 MHz
Efficiency	30–60% (drops V_{in} − V_0 across pass device as heat)	**70–95%**
Noise/ripple	Very low (clean)	Switching noise/EMI present
Size/weight	Heavy (mains transformer, heat sink)	Small (HF transformer/inductor)
Output vs input	Step-down only (V_0 < V_{in})	Buck (down), **boost (up)**, buck-boost (invert/either)
Use	Audio, precision analog	Computers, chargers, TVs

Basic SMPS topologies: **buck** ($V_0 = D \cdot V_{in}$), **boost** ($V_0 = V_{in}/(1-D)$), **buck-boost** ($V_0 = -D \cdot V_{in}/(1-D)$); D = duty cycle. Control by **PWM** of the switching transistor.

4.6 IC 555 Timer — in depth ★

8-pin monolithic timer; supply 4.5–16 V (up to 18 V); output can sink/source 200 mA; name from the **three 5 kΩ resistors** of the internal divider.

Pins: 1 GND, 2 Trigger, 3 Output, 4 Reset (active-low), 5 Control voltage, 6 Threshold, 7 Discharge, 8 $+V_{CC}$.

4.6.1 Internal block diagram ★

- **Voltage divider:** three equal 5 kΩ resistors set two references: **$(2/3)V_{CC}$** and **$(1/3)V_{CC}$**.
- **Comparator 1 (upper):** (+) = Threshold pin 6, (−) = 2/3 V_{CC}. When threshold > 2/3 V_{CC} → **R** input of flip-flop → output LOW.
- **Comparator 2 (lower):** (−) = Trigger pin 2, (+) = 1/3 V_{CC}. When trigger < 1/3 V_{CC} → **S** input → output HIGH.
- **SR flip-flop** drives the **output buffer (pin 3)** and the **discharge transistor** (open-collector NPN, pin 7) — transistor ON (capacitor discharges) when output is LOW.
- **Reset (pin 4):** overrides everything, forces output LOW when pulled < ~0.7 V; tie to V_{CC} when unused.
- **Control (pin 5):** direct access to the 2/3 V_{CC} node — external voltage here changes both thresholds ⇒ **voltage-controlled pulse width / VCO**; bypass with 0.01 µF when unused.

4.6.2 Monostable (one-shot) mode ★

R from V_{CC} to pins 6+7, C from there to ground; negative trigger pulse (< 1/3 V_{CC}) on pin 2.

Action: trigger → output HIGH, discharge transistor OFF → C charges through R toward V_{CC} → when v_C reaches $(2/3)V_{CC}$, comparator 1 resets FF → output LOW, C discharges. One clean pulse per trigger.

Derivation of T: $v_C(t) = V_{CC}(1 - e^{(-t/RC)})$. Set $v_C = (2/3)V_{CC}$:

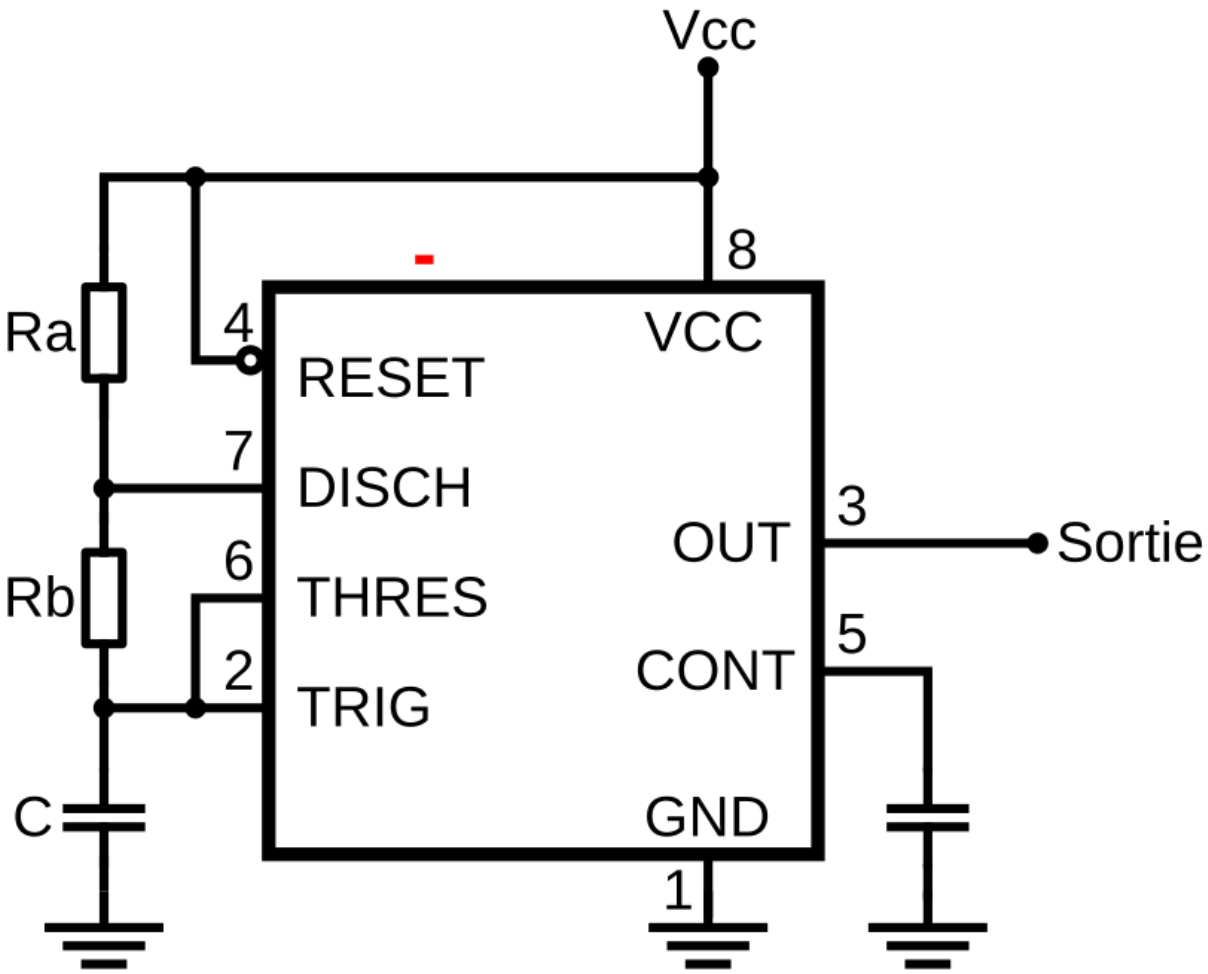
$2/3 = 1 - e^{(-T/RC)} \Rightarrow e^{(-T/RC)} = 1/3 \Rightarrow$ **$T = RC \cdot \ln 3 = 1.0986\,RC \approx 1.1\,RC$**

WE-9 (Monostable). Design a 1.1 ms one-shot with C = 0.1 µF:

$R = T/(1.1C) = 1.1\times10^{-3}/(1.1\times10^{-7}) =$ **10 kΩ**. (Check: $T = 1.1 \times 10^4 \times 10^{-7} = 1.1$ ms ✓)

Applications: timing delay, pulse-width stretching, missing-pulse detector, frequency divider, debouncing, PWM (with control-pin modulation).

4.6.3 Astable (free-running) mode ★



555 timer in astable mode.

R_A from V_{CC} to pin 7, R_B from pin 7 to pins 6+2, C from there to ground. No stable state: C charges through **R_A+R_B** from 1/3 to 2/3 V_{CC} (output HIGH), discharges through **R_B** alone via pin 7 from 2/3 to 1/3 V_{CC} (output LOW).

t_{HIGH} = 0.693 (R_A+R_B) C t_{LOW} = 0.693 R_B C

T = 0.693 (R_A+2R_B) C f = 1.44 / ((R_A+2R_B) C)

Duty cycle D = t_{HIGH}/T = (R_A+R_B)/(R_A+2R_B) — always **> 50%**; D → 50% when R_B ≫ R_A; exact 50% (and < 50%) needs a diode across R_B (charging bypasses R_B: then t_H = 0.693R_AC, t_L = 0.693R_BC).

(The 0.693 factor is ln 2 — charging between fixed fractions 1/3→2/3 of V_{CC} always takes τ·ln 2.)

WE-10 (Astable). R_A = 2.2 kΩ, R_B = 4.7 kΩ, C = 0.1 µF.

t_{HIGH} = 0.693(6.9k)(0.1µ) = 0.478 ms; t_{LOW} = 0.693(4.7k)(0.1µ) = 0.326 ms.

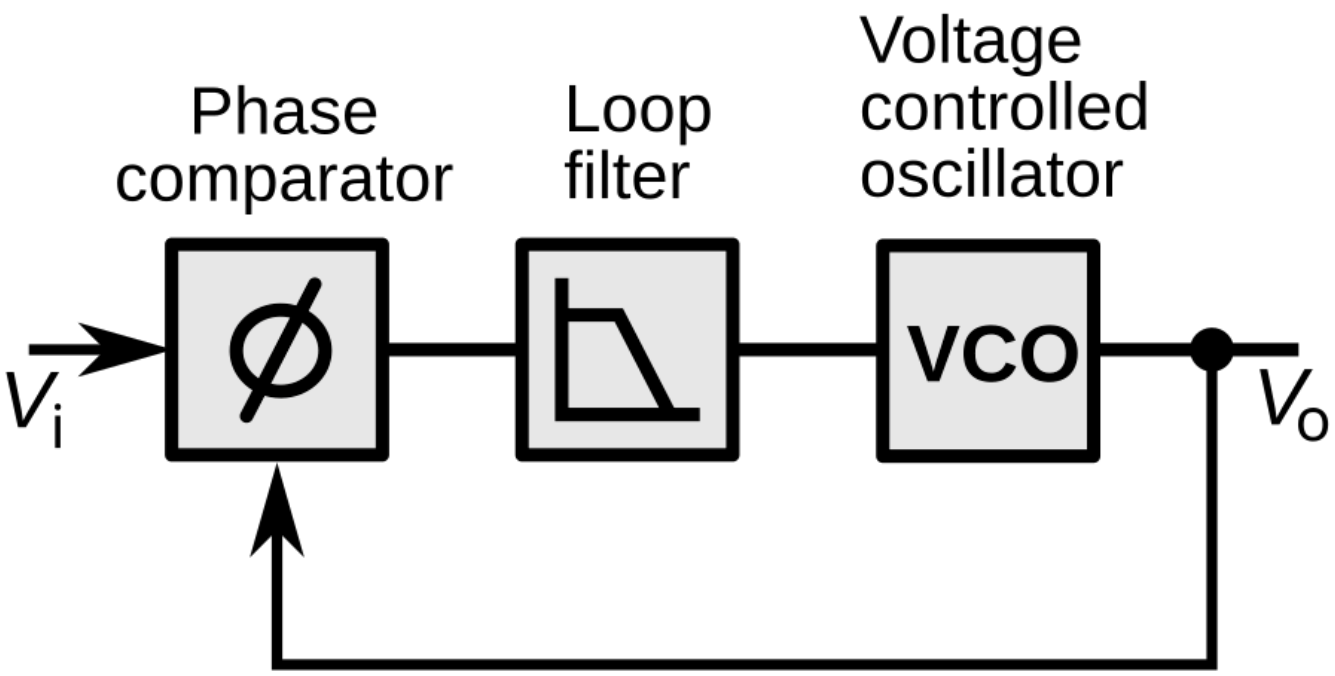
T = 0.804 ms ⇒ **f ≈ 1.24 kHz**; D = 6.9/11.6 = **59.5%**.

(Formula check: f = 1.44/((2.2k+9.4k)×0.1µ) = 1.44/(1.16×10^{-3}) ≈ 1.24 kHz ✓)

Applications: clock generation, LED/lamp flasher, tone generation, FSK modulator, voltage-controlled oscillator via pin 5, ramp generation.

556 = dual 555; **7555** = CMOS 555 (µA-level supply current, up to 2 MHz).

4.7 Phase-Locked Loop (PLL) — in depth ★ (also feeds Communication unit)



Phase-locked loop (PLL) block diagram.

A PLL is a **negative-feedback frequency/phase control system** that locks an internal oscillator's frequency *and phase* to an incoming signal.

4.7.1 Block diagram ★

```
f_in ──►[ Phase Detector / Comparator ]──►[ Low-Pass Filter ]──►[ Error Amplifier ]──┬──► V_c (control voltage)
                  ▲                                                                  │
                  └──────────────────[ VCO ]◄────────────────────────────────────────┘
                                     f_out (fed back)
```

Block	Function
Phase detector (PD)	Multiplies/compares f_{in} with VCO output; produces error voltage ∝ **phase difference** (plus sum-frequency terms)
Low-pass filter (LPF)	Removes high-frequency components of the PD output, leaves DC error; **its bandwidth sets the capture range** and loop dynamics
Error amplifier	Amplifies the DC error voltage
VCO	Free-runs at f_0 (centre frequency); output frequency shifts in proportion to control voltage V_c: f = f_0 + K_VV_c

Operation states: free-running (no input; VCO at f_0) → **capture** (loop pulling VCO toward f_{in}) → **locked/tracking** (f_{VCO} = f_{in} exactly; only a small static phase difference — ideally 90° for an analog multiplier PD — remains, generating the needed V_c).

4.7.2 Lock range vs capture range ★

	Lock (tracking) range Δf_L	Capture (acquisition) range Δf_C
Meaning	Band over which PLL, **once locked**, can *stay* locked as f_{in} drifts	Band over which an **unlocked** PLL can *acquire* lock
Determined by	PD gain, amplifier gain, VCO gain (loop DC gain)	Mainly **LPF bandwidth** (+ loop gain)
Relation	**Δf_C ≤ Δf_L** — capture range is always the smaller, both centred on f_0	

For the IC 565: lock range **$f_L = \pm 8 f_0 / V$** (V = total supply $V^+ - V^-$); capture range **$f_C = \pm\sqrt{f_L / (2\pi \cdot (3.6 \times 10^3) \cdot C)}$**, C = LPF capacitor. Pull-in time decreases with wider LPF.

4.7.3 Applications ★★ (asked under Communication in 2025)

1. **FM demodulation (most-asked):** locked PLL forces the VCO to track the FM input's instantaneous frequency; the **VCO control voltage V_c is therefore a replica of the original modulating signal — V_c itself is the demodulated output**. Linear, needs no tuned circuits — the standard IC FM detector.
2. **AM detection (synchronous/coherent):** PLL locks to the AM carrier and regenerates it; the regenerated carrier (phase-shifted 90° back to 0°) multiplies the AM signal in a separate mixer; the LPF output is the envelope. Superior noise performance vs diode envelope detector.
3. **Frequency synthesis / multiplication:** insert a **÷N counter** in the feedback path between VCO and PD ⇒ lock forces f_{VCO}/N = f_{in} ⇒ **f_{out} = N·f_{in}**. A programmable N gives many crystal-stable channels from one reference — used in every synthesised receiver/transmitter. (Divide the *input* instead ⇒ frequency division; harmonic locking also multiplies.)
4. **FSK demodulation:** FSK's two frequencies produce two distinct DC levels at the PLL error output; a comparator restores the binary data (e.g. 1070/1270 Hz modem tones).
5. **Others:** carrier recovery, clock recovery from data, bit synchronisation, tone decoding (IC 567), motor speed control, frequency translation.

4.7.4 IC 565 basics

- Monolithic PLL, 14-pin; usable to **~500 kHz**; supply ±6 V to ±12 V.

- Contains PD (analog multiplier type), error amplifier, and a highly linear VCO; the **LPF is external** (internal 3.6 kΩ resistor + external capacitor C to supply).
- **Free-running frequency: $f_0 = 1.2/(4R_1C_1) = 0.3/(R_1C_1)$** ($R_1$, C_1 = external timing components; 2 kΩ ≤ R_1 ≤ 20 kΩ recommended).
- VCO output (pin 4) is externally connected to PD input (pin 5) — the break lets you insert a ÷N counter for synthesis.
- Related ICs: **NE566** = VCO only; **NE567** = PLL tone decoder.

WE-11 (565 free-running frequency). R_1 = 10 kΩ, C_1 = 0.01 µF:

$f_0 = 0.3/(10^4 \times 10^{-8}) = 0.3/10^{-4}$ = **3 kHz**. With V_{total} = 12 V, lock range f_L = ±8(3000)/12 = **±2 kHz** (i.e. 1 kHz–5 kHz).

WE-12 (Schmitt trigger trip points). Inverting Schmitt: R_1 = 10 kΩ (to ground), R_2 = 90 kΩ (to output), V_{sat} = ±13 V.

β = 10/(10+90) = 0.1 ⇒ UTP = +1.3 V, LTP = −1.3 V, hysteresis = **2.6 V**. A noisy sine of 0.5 V ripple riding on the signal cannot false-trigger (0.5 V < 2.6 V).

4.8 PYQ Corner — actual 22-04-2025 CBT-2 questions, fully solved

PYQ 1 (2025). *In the internal block diagram of IC 741 op-amp, the second stage is:*

(a) Input differential amplifier (b) **Intermediate (gain) stage** (c) Complementary push-pull output stage (d) Class-B amplifier stage

Solution. 741 stage order: **input differential stage → intermediate/gain stage → level shifter → push-pull output**. The second stage is the **intermediate gain stage** — a dual-input *unbalanced*-output differential amplifier providing most additional gain (some texts merge it with the level shifter and call it the "level-shift and gain" stage). **Answer: (b).**

PYQ 2 (2025). *The output of an op-amp integrator is:*

(a) Proportional to the derivative of the input (b) Equal to the input (c) **Proportional to the integral of the input** (d) Proportional to the logarithm of the input

Solution. With C in feedback and R at the inverting input, the virtual ground gives $I = V_{in}/R = -C\,dV_0/dt \Rightarrow$ **$V_0 = -(1/RC)\int V_{in}\,dt$**. Output ∝ (negative) integral of input; a square-wave input yields a triangular output. **Answer: (c).**

PYQ 3 (2025, asked under Communication — PLL-framed). *In a PLL used as an FM demodulator, the demodulated output is taken from:*

(a) VCO output (b) Phase-detector direct output (c) **Error/control voltage applied to the VCO (after the LPF)** (d) The feedback divider

Solution. In lock, the VCO exactly tracks the FM signal's instantaneous frequency; the control voltage that steers the VCO must therefore vary exactly as the modulating signal did ⇒ **V_c (LPF/error-amp output) = demodulated audio**. **Answer: (c).**

PYQ 4 (2025, Communication-framed). *The capture range of a PLL is:*

(a) Always greater than the lock range (b) **Always less than or equal to the lock range and decided mainly by the LPF bandwidth** (c) Independent of the LPF (d) Equal to the VCO free-running frequency

Solution. An unlocked loop must pass the beat (difference) frequency through the LPF to develop a pull-in voltage — so a narrower LPF ⇒ smaller capture range. Once locked, the PD output is near-DC and passes regardless, so tracking (lock) extends further: **$\Delta f_C \le \Delta f_L$**, both centred on f_0. **Answer: (b).**

4.9 Practice MCQs (20)

1. In a monolithic IC, all components are fabricated on:

(a) separate substrates (b) a single silicon chip (c) a ceramic film (d) a PCB

2. The first (input) stage of IC 741 is a:

(a) push-pull amplifier (b) level shifter (c) dual-input balanced-output differential amplifier (d) class-C amplifier

3. The purpose of the level-shifting stage in 741 is to:

(a) increase gain (b) shift the DC level so the output swings about 0 V (c) increase bandwidth (d) reduce noise

4. An ideal op-amp has:

(a) zero input resistance (b) infinite output resistance (c) infinite CMRR (d) finite bandwidth

5. The slew rate of IC 741 is approximately:

(a) 0.5 V/µs (b) 5 V/µs (c) 50 V/µs (d) 0.5 V/ms

6. In an inverting amplifier, the inverting input terminal is at virtual ground because:

(a) it is wired to ground (b) A is very large with negative feedback so $V_- \approx V_+ = 0$ (c) input resistance is zero (d) output is zero

7. For R_1 = 1 kΩ, R_f = 10 kΩ, the non-inverting amplifier gain is:

(a) −10 (b) 10 (c) 11 (d) −11

8. A voltage follower is chiefly used for:

(a) high voltage gain (b) impedance matching / buffering (c) phase inversion (d) rectification

9. A square wave applied to an op-amp integrator produces:

(a) spikes (b) a sine wave (c) a triangular wave (d) a square wave

10. A triangular wave applied to an op-amp differentiator produces:

(a) a square wave (b) a triangular wave (c) a ramp (d) a sine wave

11. In a practical integrator, a resistor R_f is connected across C to:

(a) increase gain at all frequencies (b) limit the DC gain and prevent output saturation (c) make it a differentiator (d) improve slew rate

12. A zero-crossing detector converts a sine input into:

(a) a triangular wave (b) a square wave (c) a DC level (d) spikes

13. In an op-amp Schmitt trigger with V_{sat} = ±12 V, R_1 = 1 kΩ (to ground), R_2 = 5 kΩ (to output), the hysteresis is:

(a) 2 V (b) 4 V (c) 12 V (d) 1 V

14. A precision rectifier can rectify signals smaller than 0.7 V because:

(a) it uses Schottky diodes (b) the diode is inside the feedback loop, dividing its drop by the open-loop gain (c) it uses a transformer (d) the op-amp saturates

15. In a Wien-bridge oscillator using an op-amp, the minimum amplifier gain and R_f/R_1 ratio are:

(a) 3 and 2 (b) 2 and 3 (c) 29 and 28 (d) 1 and 0

16. The dropout voltage of a 7805 regulator is about:

(a) 0.2 V (b) 2 V (c) 5 V (d) 7 V

17. For an LM317 with R_1 = 240 Ω and R_2 = 1.2 kΩ, the output voltage is:

(a) 1.25 V (b) 6.25 V (c) 7.5 V (d) 5 V

18. In a 555 astable circuit, the capacitor discharges through:

(a) R_A only (b) R_A + R_B (c) R_B only (via the discharge pin) (d) the output pin

19. A 555 monostable with R = 100 kΩ and C = 10 µF gives a pulse width of about:

(a) 0.11 s (b) 1.1 s (c) 11 s (d) 1.44 s

20. In a PLL frequency synthesiser, a ÷N counter placed between the VCO and the phase detector makes the output frequency:

(a) f_{in}/N (b) N·f_{in} (c) f_{in} (d) f_0/N

Answer key with brief explanations

Q	Ans	Why
1	b	Monolithic = "single stone": everything diffused into one silicon chip
2	c	Input stage = dual-input, balanced-output diff amp (high Z_{in}, high CMRR)
3	b	Direct coupling builds up DC; the level translator restores 0 V quiescent output
4	c	Ideal: $A = \infty$, $R_{in} = \infty$, $R_{out} = 0$, BW = ∞, CMRR = ∞
5	a	741 SR ≈ 0.5 V/μs — its main speed limitation
6	b	Virtual short with (+) grounded ⇒ $V_- \approx 0$ without a physical ground
7	c	$A = 1 + R_f/R_1 = 1 + 10 = 11$
8	b	Unity gain, Z_{in} huge, $Z_{out} \approx 0$ ⇒ buffer
9	c	Integral of a constant = ramp; alternating ramps = triangle
10	a	Derivative of the triangle's constant slopes = alternating constants = square
11	b	R_f makes DC gain finite ($-R_f/R$) so offset can't integrate to saturation
12	b	Output flips between $\pm V_{sat}$ at each zero crossing
13	b	$V_H = 2V_{sat}R_1/(R_1+R_2) = 2\times12\times1/6 = 4$ V
14	b	Effective diode drop ≈ 0.7/A ≈ microvolts
15	a	Wien network feeds back 1/3 at f_0 ⇒ gain 3 ⇒ $R_f = 2R_1$
16	b	78xx needs ≈ 2 V input–output headroom
17	c	$V_0 = 1.25(1 + 1200/240) = 1.25\times6 = 7.5$ V
18	c	Discharge path: C → R_B → pin 7 transistor → ground
19	b	$T = 1.1RC = 1.1\times10^{5}\times10^{-5} = 1.1$ s
20	b	Lock forces $f_{VCO}/N = f_{in}$ ⇒ $f_{out} = N \cdot f_{in}$

4.10 Rapid-Revision Formula Sheet

Op-amp basics

- $V_0 = A(V_+ - V_-)$; 741: $A \approx 2\times10^5$, $R_{in} \approx 2$ MΩ, $R_{out} \approx 75$ Ω, SR ≈ 0.5 V/µs, GBW ≈ 1 MHz, CMRR ≈ 90 dB
- 741 stages: **Input diff amp → Intermediate gain → Level shifter → Push-pull output ★**
- CMRR (dB) = $20 \log(A_d/A_{cm}) \cdot f_{max} = SR/(2\pi V_m) \cdot BW_{CL} = GBW/A_{CL}$
- Golden rules (with −ve feedback): $V_+ = V_-$ (virtual short); input currents = 0

Linear circuits

Circuit	Output
Inverting amp	$V_0 = -(R_f/R_1)V_{in}$
Non-inverting amp	$V_0 = (1 + R_f/R_1)V_{in}$
Follower	$V_0 = V_{in}$ (buffer)
Inverting summer	$V_0 = -R_f(V_1/R_1 + V_2/R_2 + \ldots)$
Difference amp	$V_0 = (R_2/R_1)(V_2 - V_1)$
Instrumentation amp	$V_0 = (1 + 2R/R_G)(V_2 - V_1)$
Integrator ★	$V_0 = -(1/RC)\int V_{in}dt$ (square → triangle)
Differentiator	$V_0 = -RC \cdot dV_{in}/dt$ (triangle → square)
I→V converter	$V_0 = -I_{in}R_f$
V→I converter	$I_L = V_{in}/R_1$ (load-independent)

Non-linear / filters / oscillators

- Schmitt: $UTP/LTP = \pm V_{sat}R_1/(R_1+R_2)$; hysteresis $= 2V_{sat}R_1/(R_1+R_2)$
- Op-amp astable: $T = 2RC \ln((1+\beta)/(1-\beta))$; $\beta = 0.5 \Rightarrow T \approx 2.2RC$
- Wien bridge: $f_0 = 1/(2\pi RC)$; gain = 3 ($R_f = 2R_1$) · Phase-shift: $f_0 = 1/(2\pi RC\sqrt{6})$, gain 29
- Filters: $f_c = 1/(2\pi RC)$; 1st order 20 dB/dec, 2nd order 40 dB/dec; Butterworth = maximally flat (2nd-order S-K gain 1.586); $f_0(BP) = \sqrt{(f_L f_H)}$, $Q = f_0/BW$

Regulators

- 78xx = +ve fixed, 79xx = −ve fixed; dropout ≈ 2 V; pins 78xx: In-Gnd-Out; 79xx: Gnd-In-Out
- LM317: **$V_0 = 1.25(1 + R_2/R_1)$** ★; range 1.25–37 V; LM337 = negative twin
- 723: reference 7.15 V, 2–37 V adjustable, foldback limiting
- SMPS: buck $V_0 = DV_{in}$; boost $V_0 = V_{in}/(1-D)$; efficiency 70–95% vs linear 30–60%

555 timer ★

- Thresholds: trigger < $(1/3)V_{CC}$ → HIGH; threshold > $(2/3)V_{CC}$ → LOW; three 5 kΩ divider resistors
- Monostable: **$T = 1.1\ RC$**
- Astable: $t_H = 0.693(R_A + R_B)C$; $t_L = 0.693R_BC$; **$f = 1.44/((R_A+2R_B)C)$**; $D = (R_A+R_B)/(R_A+2R_B) > 50\%$

PLL ★

- Blocks: **Phase detector → LPF → Error amp → VCO** (feedback to PD)
- Capture range ≤ lock range; capture set by LPF bandwidth; both centred on f_0
- FM demodulation: **output = VCO control voltage** ★
- ÷N in feedback ⇒ f_{out} = N·f_{in} (synthesis) · FSK: two frequencies → two DC levels → comparator
- IC 565: $f_0 = 0.3/(R_1C_1) = 1.2/(4R_1C_1)$; f_L = $\pm 8f_0/V$; NE566 = VCO, NE567 = tone decoder

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End of Chapter 4. Next: Chapter 5 — Microprocessor & Microcontroller (8085/8051).